

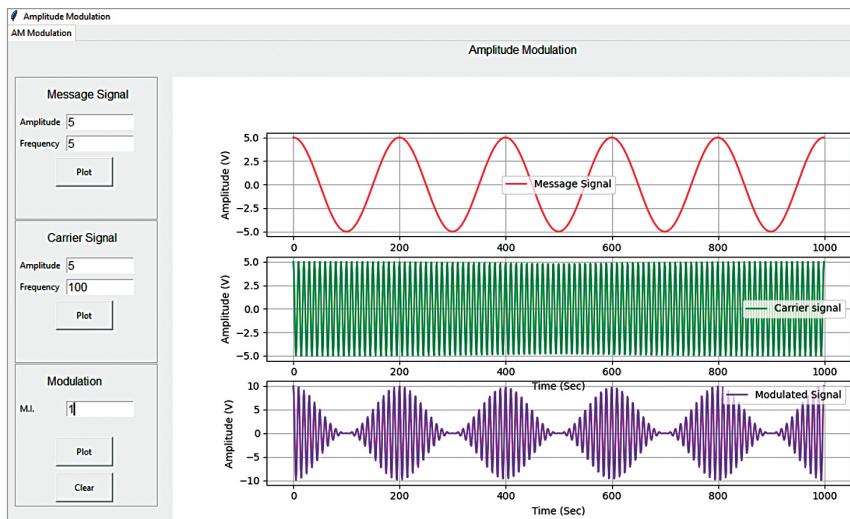


Fig. 5: Perfect modulation

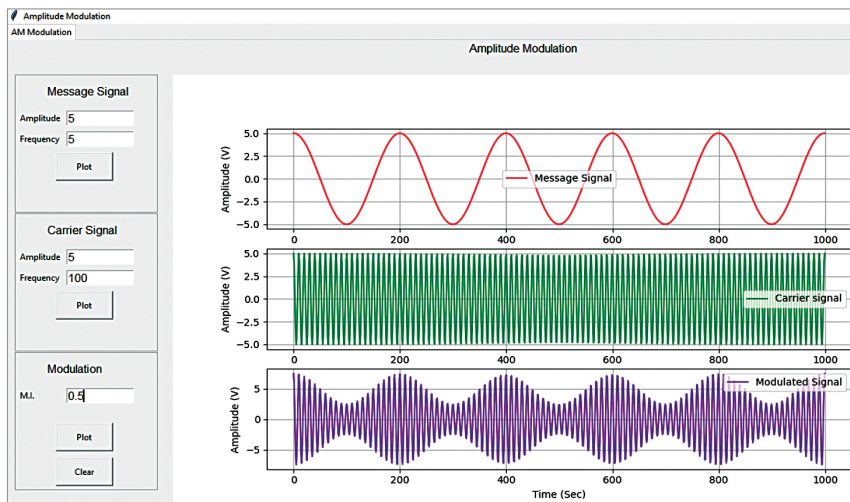


Fig. 6: Under modulation

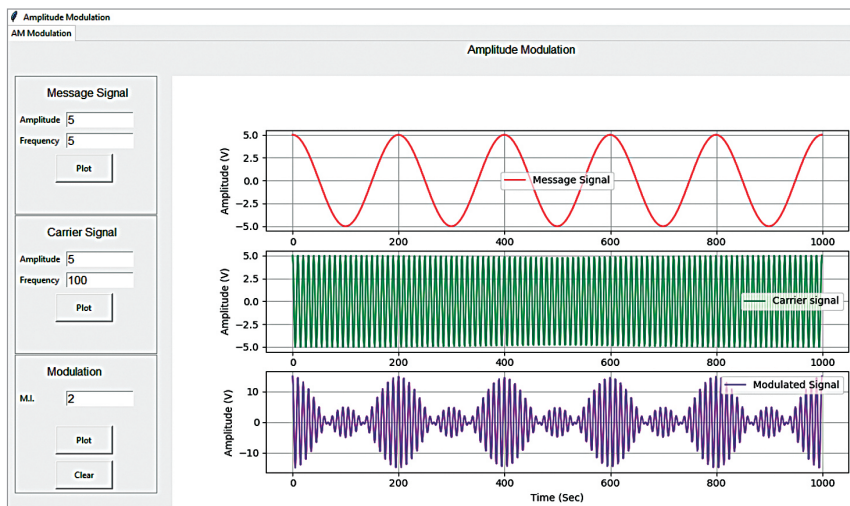


Fig. 7: Over modulation

EFY Note

The source code of this project is available for free download at www.electronicsforu.com

Two button widgets are used for plotting the message and carrier signals. The commands for these buttons are defined by taking inputs from entry widget variables (Amplitude and Frequency) and using sin function from NumPy package of Python.

The mathematical equation for modulated signal is defined in Eq 3. This is plotted by using the NumPy package. The user-defined modulation index is used for calculating and plotting the modulated signal.

A button for clearing the plotting in canvas needs to be designed. This can be achieved by `.clear()` function.

To test, run the simulator code and put the values for signal modulation. Test perfect modulation under Modulation and over modulation of signals using the simulation. The testing cases are shown as following.

When the modulation index is maintained at value 1, as shown in Fig. 5, you get perfect modulation. When modulation index is below 1, as shown in Fig. 6, it is called under modulation. When modulation index is above 1, as in Fig. 7, you get over modulation. **EFY**



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